The Jordan–Schönflies theorem: a complete proof for human readers

Abstract

We prove: every homeomorphism between two Jordan curves extends to a homeomorphism of the plane. The proof is complete — every claim is argued — but written at ordinary mathematical granularity. It is the compact companion of a longer manuscript prepared as a formalization blueprint. The two documents follow the same route and are sectioned alike, so any step here can be expanded by consulting it; where they differ, it is because an argument has been shared between two places rather than written out twice.

The plan

A Jordan curve may be monstrous. It can be nowhere differentiable, it can have positive area, and it can meet a line in a Cantor set. Nothing can be constructed on it directly, and every proof of the theorem is therefore a proof that the monstrosity is irrelevant: that the curve is controlled entirely by objects one *can* compute with. Here those objects are polygons and finite graphs.

Part I, Sections 2–5, makes them do the separating. For a polygonal curve, separation is arithmetic: a point is inside precisely when a ray to the right crosses the curve an odd number of times (Section 3). Everything else in Part I is extracted from that single count. It yields the polygonal Jordan and crosscut theorems, and through them the one nonplanarity fact we need — that $K_{3,3}$ cannot be drawn in the plane — which is the lever that reaches arbitrary curves. A Jordan curve whose complement had three regions would let us draw $K_{3,3}$; so would a curve that failed to separate at all. And a simple arc cannot separate, because a chain of small squares laid along it has a corridor for an outer face. These three facts are the Jordan curve theorem (Section 5). Observe what Part I never does: it never approximates a Jordan curve by polygons. It approximates the *paths that would have to cross it*, which is a far cheaper thing to control.

Part II, Sections 6–12, builds the extension. It never writes down a homeomorphism from the closed Jordan domain $\overline{D}$ to a square Q; it writes down a *sequence of finite combinatorial matches* and takes a limit. A *matched cellulation* is a pair of finite decompositions, one of $\overline{D}$ and one of Q, into vertices, edges and faces, together with an isomorphism between them that already agrees with the given boundary map. Such a pair can be refined on either side and the refinement copied to the other — this is the finite transfer theorem of Section 8, and it is possible only because both sides are, at every finite stage, plane graphs governed by the crosscut theorem. Alternating the two directions (Section 9) drives the cells small on both sides at once. Each point of D then sits in a nested sequence of closed stars whose partners in Q shrink to a single point, and that point is its image (Section 10).

Three seams take care. The refinements must shrink cells on *both* sides, which is why the scheme alternates a grid on the source with a mesh on the target rather than refining one side freely. The transfer is asymmetric: copying into the target is unobstructed, since everything there is polygonal, whereas copying into the source needs a supply of boundary points approachable along a straight segment — the *strongly accessible* points, which are dense (Section 6). And the limit map is built inside D, so it says nothing about the boundary; continuity there is recovered separately, in Section 11, by trapping the curve between crosscuts. Section 12 passes from the closed disc to the whole plane by an inversion.

The route is Thomassen's; the work here is to leave no step unargued.

1 Statement, conventions, and background

Theorem 1.1 (Relative Jordan–Schönflies)**.** *Let C, C' be Jordan curves in the plane and $f : C \to C'$ a homeomorphism. Then f extends to a homeomorphism of $\mathbb{R}^2$ onto itself.*

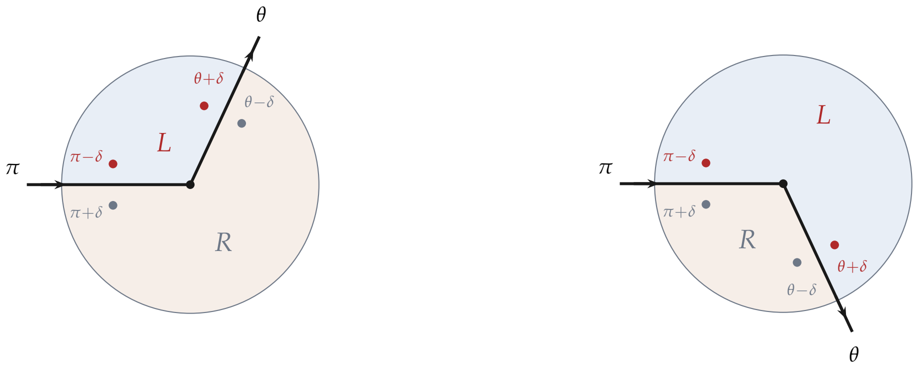
A *simple arc* is the image of an injective continuous map from $[0,1]$; a *Jordan curve* is the image of a continuous $\gamma : [0,1] \to \mathbb{R}^2$, injective on $[0,1)$, with $\gamma(0) = \gamma(1)$. An arc or curve is *polygonal* if it is a finite union of segments. A *region* is a nonempty open connected subset of the plane. Unadorned norms and distances are Euclidean; a subscript ∞ denotes the sup norm, and $\|x\|_\infty \le \|x\| \le \sqrt{2}\,\|x\|_\infty$. For a plane set E, E° is the topological interior, except that for a simple arc or a segment P we write P° for P minus its two endpoints.

We freely use standard facts of plane topology: compact = closed and bounded, sequential compactness, attainment of minima, uniform continuity on compacta, that a continuous bijection from a compact space onto a Hausdorff space is a homeomorphism, that finitely many continuous maps on closed sets agreeing on overlaps paste to a continuous map, that continuous images and unions-with-a-common-point of connected sets are connected, that a nonempty clopen subset of a connected space is everything, that components of open subsets of the plane are open, that the boundary of a component of the complement of a closed set lies in that closed set, that a continuous surjection carries dense sets to dense sets, and that $\mathbb{Q}^2$ is dense. From finite graph theory we use only that any walk between two vertices contains a simple path (shorten at repeated vertices) and that minimal objects exist by finiteness. Four small lemmas of the same character are used often enough to state.

Lemma 1.2 (Compact separation)**.** (a) *A compact set contained in an open set U has an open ρ-neighborhood contained in U for some $\rho > 0$.* (b) *Two disjoint nonempty compact sets are at positive distance.* (c) *If $x \in U \setminus K$ with U open and K compact, some ball around x lies in $U \setminus K$.*

Proof. (a) $\operatorname{dist}(\cdot, \mathbb{R}^2 \setminus U)$ has a positive minimum on the compact set (or the complement is empty). (b) Minimize distance on the compact product. (c) Take a ball inside U and shrink it below $\operatorname{dist}(x, K) > 0$. □

Lemma 1.3 (Nested compact singleton)**.** *If $K_0 \supseteq K_1 \supseteq \cdots$ are nonempty compact plane sets with* $\operatorname{diam} K_n \to 0$, *then $\bigcap_n K_n$ is a single point.*



(a) a left turn

(b) a right turn

Figure 1: The bookkeeping at a vertex. With the incoming tangent at angle 0, the two left germs (red) sit at polar angles $\pi - \delta$ and $\theta + \delta$, the two right germs (grey) at $\pi + \delta$ and $\theta - \delta$. Whatever the sign of the turn, the left pair lies in the counterclockwise arc from θ to π — shaded L — and the right pair in the complementary arc. This is why the two tracks of a strip never change places, and why they close up after a full circuit.

Proof. Pick $x_n \in K_n$; a subsequence converges to some x, which lies in every closed K_m since the tail of the subsequence does. Two points of the intersection are at distance $\leq \operatorname{diam} K_n \to 0$. □

Lemma 1.4 (Recognizing a component)**.** *Let V be an open plane set and $U \subseteq V$ a region with $\partial U \cap V = \varnothing$. Then U is a component of V.*

Proof. U is open, and closed in V because $\overline{U} \cap V = (U \cup \partial U) \cap V = U$. Thus U is a nonempty clopen connected subset of the component of V containing it, hence is that component. □

Lemma 1.5 (Polygonal connectedness)**.** *Any two points of a region are joined in it by a simple polygonal arc. The same holds in any connected set that is a finite union of polygonal arcs.*

Proof. The set of points joinable to a fixed one by polygonal paths is open (a disk in the region is convex) and has open complement in the region, hence is everything. To make the path simple, subdivide its segments at all mutual intersections and endpoints of overlaps, obtaining a finite graph, and take a simple graph path. The second statement follows by the same subdivision. □

2 Two-sided strips along polygonal curves

That a polygonal curve has two sides is the first thing Part I needs and the last thing it can take for granted. The following lemma produces the two sides explicitly, as a neighborhood split into two connected tracks. It is what bounds the number of components of a polygonal curve's complement, and it returns much later to bound the number of sides of a crosscut (Lemma 6.1).

Lemma 2.1 (Two-sided strips)**.** *Let P be a simple closed polygonal curve, or a simple polygonal arc with a compact subarc K of $P \setminus \partial P$. Then P (respectively K) has an open neighborhood N, inside any prescribed open set containing it, with $N \setminus P = N_L \sqcup N_R$, both open and connected. Moreover, for every $x \in P$ (resp. $x \in K$) and every sufficiently small disk B at x, the set $B \setminus P$ has exactly two components, one in N_L and one in N_R.*

Proof. Delete vertices at which consecutive edges are collinear, and orient P. Around each remaining vertex take a small closed disk; around each edge a thin rectangular block, extended slightly into its two vertex disks. By Lemma 1.2, all these can be chosen so that only consecutive blocks overlap, and only in small rectangles around the radial entry segments; all closures of nonadjacent blocks are disjoint, and no block meets a nonincident edge.

Call the side of a directed edge obtained by rotating its tangent counterclockwise through $\pi/2$ its *left*. At a vertex, put the incoming tangent at angle 0, so the ray back along the incoming edge has angle π, and let the outgoing tangent have angle θ. Left-side points of the incoming and outgoing strips near the vertex have polar angles $\pi - \delta$ and $\theta + \delta$ (Figure 1) for small $\delta > 0$; both lie in the counterclockwise circular arc from θ to π, hence in the same component of the circle with the two directions θ, π removed, for either sign of the turn. The right-side angles $\pi + \delta$, $\theta - \delta$ lie in the other component. The degenerate values are impossible: $\theta = 0$ is a deleted collinear vertex, and $\theta = \pi$ would make the two incident segments overlap, contradicting simplicity.

Take the strips narrow enough that this matching holds inside every vertex disk, and let N be the union of the open blocks. Label each edge strip minus its core left/right, and in each vertex disk minus its two incident rays label the sector joining the two left half-strips left, the other right. Overlaps occur only in the labeled half-strips, consistently by the angle computation, so every point of $N \setminus P$ gets one label, the labeled sets are open, and each is connected because consecutive pieces overlap in nonempty half-strips. For a closed curve the labels close up consistently after one circuit, by the same computation at every vertex. For an arc, run the construction along a slightly larger compact subarc K' and cut the two extreme strips by cross-sections beyond K; no cap joining the tracks is included, and the truncated N still contains a neighborhood of each point of K.

Finally, a small disk B at $x \in P$ inside the blocks through x meets P in a diameter, a bent diameter, or the two edge germs at a vertex; in each case $B \setminus P$ has two components, each connected and hence carrying one label, and the two carry different labels by the sector/half-strip labeling. □

3 The polygonal Jordan and crosscut theorems

Let C be a simple closed polygonal curve; rotate coordinates so no edge is horizontal. For $q = (\xi, \eta) \notin C$, let $\pi_C(q) \in \{0, 1\}$ be the parity of the number of edges, oriented upward, with $a_y \leq \eta < b_y$ whose crossing of the line $y = \eta$ lies strictly right of q.

Lemma 3.1 (Crossing parity)**.** *π_C is locally constant on $\mathbb{R}^2 \setminus C$, and the two sides of a small disk crossing one edge have opposite parity.*

Proof. Away from vertex heights, the active edges and the left-right order of their crossings vary continuously, so the count changes only when q crosses an edge. At a vertex height, pass the height across each vertex right of q: a through-vertex swaps one active edge for another, a local minimum or maximum activates or retires two at once; vertices left of q never contribute. All changes are even. For the last claim, take (Lemma 1.2(c)) a disk around an interior edge point meeting no other part of C; a short horizontal segment across the edge changes the count by exactly one. $\square$

Theorem 3.2 (Polygonal Jordan curve theorem)**.** *$\mathbb{R}^2 \setminus C$ has exactly two regions, one bounded and one unbounded, both with boundary C.*

Proof. Take a strip $N_L \sqcup C \sqcup N_R$ (Lemma 2.1) and reference points z_L, z_R in the two components of a small disk at an interior edge point. For $x \notin C$, the segment from x to a nearest point $a \in C$ avoids C before a (a closer point would contradict minimality), and its final portion lies in one connected track; hence x is in the component of z_L or of z_R: at most two components. The reference points have opposite parity (Lemma 3.1), and parity is constant on connected sets, so there are exactly two components. The complement of a large disk is connected, so exactly one component is unbounded. Every point of C is a limit of both tracks, hence of both components, so $C \subseteq \partial\Omega$ for each region Ω; conversely $\partial\Omega \subseteq C$ always. Since the count vanishes far to the right, π_C is 0 on the unbounded region $\mathrm{Ext}(C)$ and 1 on the bounded region $\mathrm{Int}(C)$. $\square$

Call a Jordan curve *separating* if its complement has exactly two regions — a bounded $\mathrm{Int}(C)$ and an unbounded $\mathrm{Ext}(C)$ — each with boundary all of C. Theorem 3.2 says that every polygonal Jordan curve is separating; Theorem 5.5 will delete the word "polygonal". The next two lemmas are needed on both sides of that divide, so they are proved from the word *separating* alone, and never used before it has been established for the curves at hand.

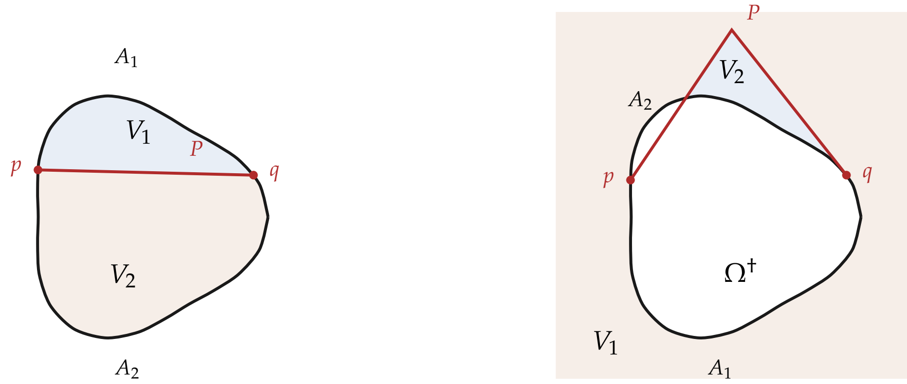
A *crosscut* of a region Ω is a simple arc with two distinct endpoints on $\partial\Omega$ and all other points in Ω. If C is separating and P is a simple arc meeting C exactly in its two endpoints, then P° is connected and misses C, so P is a crosscut of exactly one of $\mathrm{Int}(C), \mathrm{Ext}(C)$.

Lemma 3.3 (Absorption)**.** *Let C and J be separating Jordan curves, Ω a region of $\mathbb{R}^2 \setminus C$, and V a region of $\mathbb{R}^2 \setminus J$ with $\Omega \subseteq V$. Then $C \setminus J \subseteq V$.*

Proof. $C = \partial\Omega \subseteq \overline{\Omega} \subseteq \overline{V} = V \cup J$. $\square$

Lemma 3.4 (Crosscut cells)**.** *Let C be a separating Jordan curve and P a simple arc meeting C exactly in its two endpoints $p \neq q$. Let A_1, A_2 be the two arcs of C from p to q, so that $J_i = A_i \cup P$ are Jordan curves, and assume these separating as well. Let Ω be the region of $\mathbb{R}^2 \setminus C$ that P crosses and $\Omega^\dagger$ the other one, and let V_i be the region of $\mathbb{R}^2 \setminus J_i$ that does* not *contain $\Omega^\dagger$. Then V_1 and V_2 are two distinct components of $\Omega \setminus P$, and $\overline{V_i} \cap C = A_i$. (See Figure 2.)*

Proof. Each J_i is a Jordan curve because A_i and P are simple arcs meeting exactly in their two common endpoints. The region $\Omega^\dagger$ misses C and P, hence misses J_i, and being connected it lies in one region W_i of $\mathbb{R}^2 \setminus J_i$; so V_i, the other one, is well defined. Absorption applied to $\Omega^\dagger \subseteq W_i$ gives $C \setminus J_i \subseteq W_i$, which is disjoint from V_i; and $C \cap J_i =$



(a) P crosses $\Omega = \mathrm{Int}(C)$ (b) P crosses $\Omega = \mathrm{Ext}(C)$

Figure 2: A crosscut cuts two Jordan regions out of the side it crosses. In (a) the two cells are the interiors $\mathrm{Int}(J_1), \mathrm{Int}(J_2)$; in (b) — where $\Omega^\dagger = \mathrm{Int}(C)$ lies inside J_1 — the cell V_1 is the *exterior* of J_1, and only V_2 is an interior. Stating the lemma through "the region of $\mathbb{R}^2 \setminus J_i$ that does not contain $\Omega^\dagger$" covers both pictures at once. Throughout, $\overline{V_i} \cap C = A_i$.

A_i is disjoint from V_i too. So V_i is a region missing C and missing $\Omega^\dagger$, whence $V_i \subseteq \Omega$; and V_i misses $P \subseteq J_i$. Thus $V_i \subseteq \Omega \setminus P$, while $\partial V_i = J_i \subseteq C \cup P$ misses $\Omega \setminus P$, so V_i is a component of it (Lemma 1.4). The two are distinct because their boundaries are: $A_1 = A_2$ would force $C = A_1 \cap A_2 = \{p, q\}$. Finally $\overline{V_i} \cap C = (V_i \cup J_i) \cap C = A_i$. □

That is everything separation alone yields; the count is completed, in the polygonal case, by the parity function.

Lemma 3.5 (Parity splits)**.** *Let C be a simple closed polygonal curve, P a simple polygonal arc meeting C exactly in its two endpoints, and J_1, J_2 as above, with coordinates rotated so that no edge of $C \cup P$ is horizontal. Then*

$$\pi_C = \pi_{J_1} + \pi_{J_2} \pmod 2 \tag{1}$$

at every point off $C \cup P$. In particular a point of $\mathrm{Int}(C)$ *off P lies inside exactly one of J_1, J_2, and a point of* $\mathrm{Ext}(C)$ *off P lies inside both or neither.*

Proof. Subdivide C at the endpoints of P and count edge by edge: the edges of P occur in both J_i and cancel, while A_1, A_2 partition the edges of C. The two consequences restate (1) using $\pi_C = 1$ on $\mathrm{Int}(C)$ and $\pi_C = 0$ on $\mathrm{Ext}(C)$ (Theorem 3.2). □

Theorem 3.6 (Polygonal crosscuts)**.** *In the situation of Lemma 3.5, $\Omega \setminus P$ has exactly two regions, namely V_1 and V_2; and $\mathbb{R}^2 \setminus (C \cup P)$ has exactly three, namely $V_1, V_2, \Omega^\dagger$, with boundaries J_1, J_2, C.*

Proof. By Lemma 3.4 the V_i are two distinct components of $\Omega \setminus P$, so only exhaustion is at issue, and it is read off Lemma 3.5. If $\Omega = \mathrm{Int}(C)$ then $\Omega^\dagger = \mathrm{Ext}(C)$ lies in $\mathrm{Ext}(J_i)$ for both i — being connected, unbounded, and disjoint from J_i — so $V_i = \mathrm{Int}(J_i)$, and each point of $\mathrm{Int}(C) \setminus P$ lies inside exactly one J_i, that is, in exactly one V_i. If instead $\Omega = \mathrm{Ext}(C)$, then $\Omega^\dagger = \mathrm{Int}(C)$ lies inside exactly one J_i, say J_1, so $V_1 = \mathrm{Ext}(J_1)$ and

$V_2 = \mathrm{Int}(J_2)$; and a point of $\mathrm{Ext}(C) \setminus P$ lies inside both J_i — hence in V_2 — or inside neither — hence in V_1. Finally $\Omega^\dagger$ is a region disjoint from $C \cup P$ whose boundary C misses $\mathbb{R}^2 \setminus (C \cup P)$, so it is the third component (Lemma 1.4). □

Corollary 3.7 (Alternating crosscuts meet)**.** *If two polygonal crosscuts of the same side of C have four distinct endpoints alternating around C, they intersect.*

Proof. The first crosscut splits its side into V_1, V_2, whose closures meet C in the two complementary arcs A_1, A_2 between its endpoints. If the second crosscut were disjoint from the first, its connected interior would lie in one V_i and both its endpoints in $\overline{V_i} \cap C = A_i$; but alternation puts them in the relative interiors of different arcs. □

4 Plane graphs and $K_{3,3}$

A finite plane graph has distinct vertex points and simple arc edges with pairwise disjoint interiors avoiding all vertices; loops are forbidden, parallel edges allowed. Its *faces* are the components of the complement; one is unbounded. A graph is *2-connected* if it has at least three vertices, is connected, and stays connected after deleting any vertex. A *cycle* is a connected subgraph in which every vertex has degree two; a two-vertex cycle consists of two parallel edges. The realization of a cycle in a plane graph is a Jordan curve, since its edge arcs concatenate injectively.

Lemma 4.1. (a) *An edge lies on a cycle iff deleting it does not disconnect its endpoints.* (b) *A finite tree with exactly three leaves has exactly one vertex of degree ≥ 3, of degree exactly 3; all other vertices are leaves or have degree 2.* (c) *If two finite 2-connected graphs share two vertices, their union is 2-connected.* (d) *Subdividing edges of a 2-connected graph, or attaching a path (an* ear*) with distinct endpoints in the graph and new internal vertices, preserves 2-connectivity.* (e) *If $H \subseteq G$ are finite 2-connected graphs, G is obtained from H by adding ears.* (f) *The union of finitely many polygonal plane graphs becomes a plane graph after subdividing at all intersections and identifying duplicated subsegments.*

Proof. (a) One direction is clear; conversely a shortest connecting walk avoiding the edge is a simple path, which together with the edge is a cycle (possibly a two-vertex cycle). (b) A tree with n vertices has $n-1$ edges (an endpoint of a maximal simple path is a leaf; delete and induct), so $\sum_v(\deg v - 2) = -2$; three leaves contribute -3, the rest ≥ 0, forcing one vertex with $\deg -2 = 1$ and no more. (c) Deleting one vertex leaves both graphs connected and sharing a vertex. (d) For a subdivision of uv by w: deleting $x \neq w$ leaves the old connected graph with w attached to u or v; deleting w removes the edge uv, and a 2-connected graph has no bridge (a bridge's deletion would leave a component with a second vertex, whose indicated endpoint would then be a cut vertex). For an ear: deleting an old vertex leaves the old graph connected with the pieces of the ear attached at the ear's endpoints; deleting a new vertex leaves the old graph intact with two attached ear pieces. (e) Given the part K built so far, a component L of $G - V(K)$ has two distinct neighbors in K (a unique neighbor would be a cut vertex, since K has other vertices and all L-to-K paths pass through it); a shortest path through L between two neighbors is an ear. When all vertices are present, add missing edges as ears. (f)

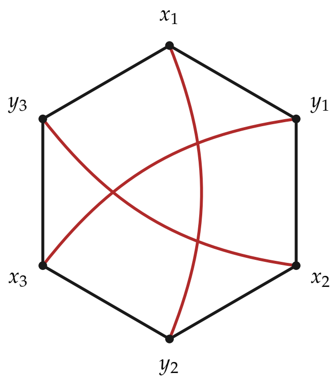


Figure 3: $K_{3,3}$ as a hexagon with three chords. Read around the hexagon, the chords join positions $(1,4)$, $(3,6)$ and $(5,2)$, so *any two of them alternate*. In a plane drawing two of the three would have to lie on the same side of the hexagon, where Corollary 3.7 forces them to meet.

Segments intersect in nothing, a point, or an interval; subdivide at all such points and interval endpoints. □

Lemma 4.2 (Polygonal redrawing)**.** *Every finite plane graph is isomorphic to one with polygonal edges.*

Proof. Take small disjoint closed squares D_v around the vertices, meeting no nonincident edge. For an edge from v to w, let $c_{v,e}$ be its last point in D_v and $c_{w,e}$ the first subsequent point in D_w; both lie on the square boundaries, and the *core* between them avoids all square interiors and all other squares. Distinct cores are disjoint compacta, with distinct boundary marks $c_{v,e}$. In $M = \mathbb{R}^2 \setminus \bigcup_v D_v^\circ$, which is locally polygonally connected (disks, half-disks along sides, three-quarter disks at corners), Lemma 1.2 yields pairwise disjoint connected relatively open neighborhoods $U_e \supseteq K_e$ meeting the square boundaries only in small disjoint arcs around the marks: the boundary minus such an arc is compact and disjoint from K_e. The clopen argument of Lemma 1.5 runs inside each U_e, so $c_{v,e}$ and $c_{w,e}$ are joined in U_e by a simple polygonal replacement core. Finally join v radially to each $c_{v,e}$ inside the convex square; radial segments meet the square boundary only at their marks, hence meet replacement cores only there, and distinct radials meet only at v. The assembled polygonal arcs realize the same abstract graph. □

Lemma 4.3. *No subdivision of $K_{3,3}$ has a plane drawing.*

Proof. In a drawing of a subdivision, each branch path is a simple arc and distinct branch paths share at most branch vertices, so a drawing of $K_{3,3}$ itself results; make it polygonal (Lemma 4.2). View $K_{3,3}$ as the hexagon $x_1y_1x_2y_2x_3y_3$ plus the chords x_1y_2, x_2y_3, x_3y_1 (Figure 3). Each chord lies on one side of the hexagon; two lie on the same side; and each pair of chords has alternating endpoints (positions $(1,4)$, $(3,6)$, $(5,2)$ pairwise alternate), contradicting Corollary 3.7. □

Lemma 4.4 (Face cycles)**.** *Every face of a finite 2-connected polygonal plane graph is one of* *the two regions of a cycle of the graph, its boundary cycle; bounded faces are interiors of their boundary cycles.*

Proof. Every vertex has two distinct neighbors (a sole neighbor would be a cut vertex), so an endpoint of a maximal simple path closes a cycle of length ≥ 3. Build the graph from that cycle by ears (Lemma 4.1(e)) and induct: an ear's interior is connected and disjoint from the current graph, so it lies in one current face F, one side of its boundary cycle; the ear's endpoints, limits of its interior, lie on ∂F; so the ear is a crosscut, and Theorem 3.6 replaces F by two regions bounded by cycles (possibly two-vertex cycles), leaving other faces untouched. $\square$

Lemma 4.5 (Outer face of a chain)**.** *Let $\Gamma_1, \ldots, \Gamma_k$ $(k \geq 3)$ be finite 2-connected polygonal plane graphs, consecutive ones sharing at least two vertices (after subdividing common points), nonconsecutive ones disjoint. A point lying in the outer face of every consecutive union $\Gamma_i \cup \Gamma_{i+1}$ lies in the outer face of the whole union G.*

Proof. G is 2-connected by Lemma 4.1(c). If x lay in a bounded face, some cycle of G would enclose it (Lemma 4.4). Choose $i \leq j$ with $j - i$ minimal so that a cycle in $\Gamma_i \cup \cdots \cup \Gamma_j$ encloses x. If $j - i \leq 1$, a consecutive pair H contains that union; the face of H containing x is connected, disjoint from the cycle, and meets its interior, hence lies inside it and is bounded — contradicting the hypothesis. So $j - i \geq 2$; among enclosing cycles in this fixed union pick C with fewest edges outside Γ_{j-1}. Interval minimality forces on C an edge of Γ_j outside Γ_{j-1} and an earlier-chain edge outside Γ_{j-1} (else shrink the interval; note Γ_j is disjoint from all Γ_ℓ, $\ell \leq j-2$). Pick interior points a, b of those two edges; each of the two arcs of C from a to b must meet Γ_{j-1}, since consecutive C-edges of the two outside types cannot share a vertex. Hence $C \cap \Gamma_{j-1}$ has at least two maximal subpaths, one per arc, and distinct maximal subpaths are vertex-disjoint. Let R be a shortest path in the connected graph Γ_{j-1} from one such subpath P to $V(C) \setminus V(P)$: its internal vertices avoid C (a C-vertex inside R would shorten it), its endpoints u, v lie on different maximal subpaths, and its edges are not C-edges (a single C-edge from u would stay inside P by maximality). Then R is a geometric crosscut of one side of C; each of the two arcs C_1, C_2 of C between u, v contains an edge outside Γ_{j-1}, since its ends lie on different maximal subpaths. Since x lies inside C and off R, Lemma 3.5 says that exactly one of the cycles $R \cup C_1$, $R \cup C_2$ encloses x; it lies in the same union and has strictly fewer edges outside Γ_{j-1}, a contradiction. So no bounded face contains x. $\square$

5 Arc complements and the Jordan curve theorem

Two statements separate the polygonal theory from the general one: an arc separates nothing, and a curve separates into exactly two. The first is proved by burying the arc in a chain of small squares and appealing to Lemma 4.5; the second, in both directions, by exhibiting a plane $K_{3,3}$.

Theorem 5.1 (Arc-complement theorem)**.** *The complement of a simple arc A is connected.*

Proof. Given $p, q \notin A$, choose $d > 0$ so that both points are at ℓ^∞-distance $> 4d$ from

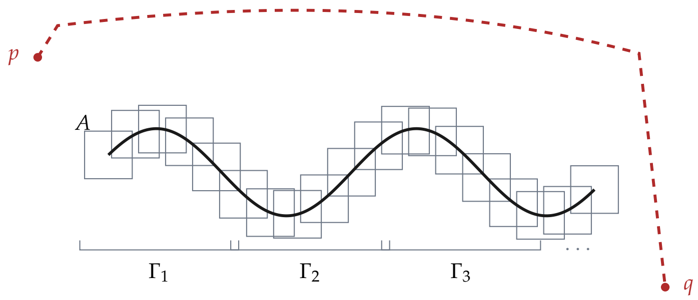


Figure 4: Why an arc cannot separate. Small squares of radius $\delta/4$ are strung along A so that consecutive ones overlap; blocks of them form 2-connected graphs Γ_i, consecutive blocks sharing a whole square and non-consecutive ones disjoint. The chain lemma says the outer face of the union is a single region — a corridor running all the way around — and it contains both p and q. The arc itself is buried: each of its points lies inside or on one of the squares.

A. By uniform continuity split $A = A_1 \cup \cdots \cup A_k$ ($k \geq 3$) into subarcs of ℓ^∞-diameter $< d$ from their initial points a_i, and let $d' > 0$ be the least distance between nonadjacent subarcs (Lemma 1.2(b)); put $\delta = \min(d, d')$. Sample each A_i so finely that every point of the subarc between two consecutive samples is within $\delta/4$ of the earlier one (hence in particular consecutive samples are within $\delta/4$), taking the common endpoint of adjacent subarcs as a sample for both, and around every sample draw the boundary of the ℓ^∞-square of radius $\delta/4$; let Γ_i be the overlay of the A_i-squares. Two consecutive squares are distinct congruent overlapping squares or coincide; in the first case each boundary has points inside and outside the other square (each center lies in the other's interior, while containment of one boundary would force, by convexity, containment of the whole square), so the boundaries meet, and in at least two points, since one shared point would disconnect a closed curve into its nonempty inside and outside parts. Hence each Γ_i is 2-connected by Lemma 4.1(c,d), and consecutive Γ_i share a whole square, while for $|i-j| \geq 2$ the graphs are disjoint (δ apart minus radii $\delta/2$). Each pair $\Gamma_i \cup \Gamma_{i+1}$ lies in the square of radius $3d$ about a_i (its centers are within d or within $2d$ of a_i, and the square radius is $\delta/4 \leq d/4$), so p, q lie in the outer face of every pair (Figure 4), hence of $G = \bigcup \Gamma_i$ (Lemma 4.5). Every point of A lies in or on a small square; interior points are separated from the outer face by the square's boundary cycle, boundary points lie on G. So the outer face, a region, avoids A and joins p to q (Lemma 1.5). $\square$

Lemma 5.2 (Circular parametrization)**.** *A Jordan curve C is homeomorphic to the unit circle (and to the boundary of a square); any two of its points split it into two arcs meeting only there, and its open subarcs are a basis of its topology.*

Proof. Let $e(t) = (\cos 2\pi t, \sin 2\pi t)$ and set $h(e(t)) = \gamma(t)$: well defined and bijective. If $z_n \to z$, lifts t_n have, along any subsequence, a further subsequence converging to some t with $e(t) = z$, along which $h(z_n) \to h(z)$; hence h is continuous, and being a continuous bijection from a compactum to a Hausdorff space, a homeomorphism. The remaining claims transport from the circle; the square case follows via radial projection. $\square$

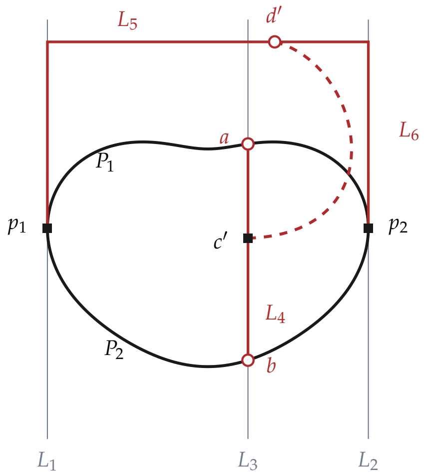


Figure 5: The configuration behind "a Jordan curve separates". Squares mark p_1, p_2, c'; circles mark a, b, d'. Every square is joined to every circle by one of nine internally disjoint arcs — four subarcs of C, two halves of L_4, two pieces of L_5, and L_6 — so the picture is a plane subdivision of $K_{3,3}$, which cannot exist. The dashed L_6 is the hypothesis being refuted — it is assumed to reach L_5 from L_4 without meeting C — so no drawing of it can be faithful; here it is shown crossing C on the right.

Proposition 5.3 (A Jordan curve separates)**.** *$\mathbb{R}^2 \setminus C$ is disconnected.*

Proof. C is not contained in a line: it would be a nondegenerate compact connected subset of it, a closed interval, which one interior point disconnects, while the circle minus a point is connected. Rotate so the horizontal width is positive; let L_1, L_2 be the vertical supporting lines, p_i the highest point of $C \cap L_i$, and P_1, P_2 the two arcs of C between p_1, p_2. A vertical line L_3 strictly between meets both arcs (their horizontal projections are connected and contain both extremes). The compact sets $P_i \cap L_3$ are disjoint (the p_i are off L_3); choose a closest pair $a \in P_1$, $b \in P_2$ on L_3; the open segment between them misses C, since a C-point on it, lying on some P_i, would be closer to the other endpoint. Call the closed segment L_4. Join p_1 to p_2 by the polygonal path L_5 running up the supporting lines and across a horizontal line above C; each interior point of L_5 reaches infinity by a horizontal ray away from the strip or a vertical ray upward, missing C, so the interior of L_5 lies in the unbounded component; and $L_4 \cap L_5 = \varnothing$.

Suppose $L_4 \setminus \{a, b\}$ also lay in the unbounded component. Join an interior point c of L_4 to an interior point d of L_5 by a simple polygonal arc R in that component (Lemma 1.5); let c' be the last point of R on L_4 and d' its first subsequent point on L_5, and L_6 the subarc between, meeting $L_4 \cup L_5$ only at its ends and missing C. Then $C \cup L_4 \cup L_5 \cup L_6$ is a plane subdivision of $K_{3,3}$ with parts $\{p_1, p_2, c'\}$ and $\{a, b, d'\}$ (Figure 5): the four subarcs of P_1, P_2 cut at a, b, the two subsegments of L_4 cut at c', the two subpaths of L_5 cut at d', and L_6 are nine internally disjoint branch paths (all the required disjointness is by construction: C meets L_4 in $\{a, b\}$, L_5 in $\{p_1, p_2\}$, L_6 not at all, and the six branch vertices are distinct since c', d' are off C). This contradicts Lemma 4.3; so some point of L_4 lies outside the unbounded component, and the complement is disconnected. $\square$

Lemma 5.4 (Accessible points are dense)**.** *Let Ω be a component of $\mathbb{R}^2 \setminus C$ and $x \in \Omega$. The*

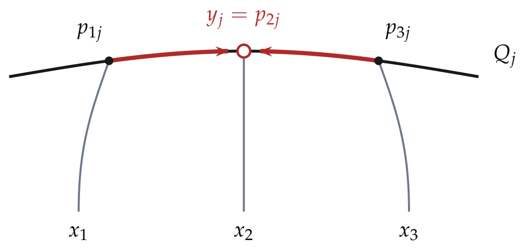


Figure 6: The middle-terminal trick. On the arc Q_j the three trees land at p_{1j}, p_{2j}, p_{3j}; the meeting point y_j is chosen to be the *middle* one. The two travelling subarcs (thick) then reach y_j from opposite sides and share no interior point, while the tree that lands on y_j travels not at all. Choosing any other terminal as y_j would make two of the three routes overlap.

points of C reachable from x by a polygonal arc meeting C only at its endpoint are dense in C.

Proof. Take y in another component and an open arc J_0 with closure inside the given open arc J of C. Then $C \setminus J_0$ is a simple arc, so its complement is connected (Theorem 5.1); join x to y there by a simple polygonal arc. The arc must cross C, and only inside J_0; its first meeting, from x, is the endpoint of an access arc from Ω and lies in J. □

Theorem 5.5 (Jordan curve theorem)**.** $\mathbb{R}^2 \setminus C$ *has exactly two regions, one bounded and one unbounded, both with boundary C.*

Proof. At least two components exist (Proposition 5.3). Suppose $\Omega_1, \Omega_2, \Omega_3$ were three. Split C into three closed arcs Q_1, Q_2, Q_3 with disjoint relative interiors, pick $q_i \in \Omega_i$, and choose nine distinct points p_{ij} in the relative interiors of the Q_j, each accessible from q_i (Lemma 5.4; a dense set meets every open subarc even after finitely many deletions, since otherwise a nonempty open subarc would avoid it). Overlay the three access arcs from q_i and take a minimal connected subgraph T_i spanning p_{i1}, p_{i2}, p_{i3}: a tree (a cycle edge could be deleted) in which each terminal has degree one, because other access arcs of q_i meet C only at their own distinct endpoints; minimality forbids other leaves. By Lemma 4.1(b), T_i is three internally disjoint branches from a center x_i to the terminals. The T_i are pairwise disjoint: their interiors lie in distinct components and the nine terminals are distinct. On each Q_j let y_j be the middle of the three terminals (Figure 6), and join each x_i to each y_j by its branch to p_{ij} followed by the subarc of Q_j to y_j (empty for the middle terminal). These nine paths are internally disjoint: branches within one tree are; trees are disjoint; the two nonempty Q_j-subarcs approach y_j from opposite sides; and subarcs lie in the disjoint relative interiors of the Q_j, which the branches meet only at their own terminals. This is a plane subdivision of $K_{3,3}$ with parts $\{x_i\}$, $\{y_j\}$, contradicting Lemma 4.3. So there are exactly two components; accessibility from both makes every point of C a boundary point of both, boundaries lie in C, and exactly one component is unbounded. □

Every Jordan curve is therefore separating, and the two lemmas of Section 3 that were proved from that word alone — absorption and crosscut cells — are from now on available for all of them.

6 Crosscuts of Jordan domains; reduction to the square

From here C is an arbitrary Jordan curve and $D = \mathrm{Int}(C)$. The crosscut theorem below is the instrument Part II uses at every single step; after it, the main theorem is reduced to the case of a square, and the rest of the paper is the construction that handles that case.

Lemma 6.1 (At most two sides)**.** *If P is a polygonal crosscut of $D = \mathrm{Int}(C)$, then $D \setminus P$ has at most two components.*

Proof. Fix z_0 interior to a straight edge of P and, by Lemma 1.2(c), a closed disk $B \subseteq D$ meeting P in a straight diameter; let z_L, z_R be points of its two half-disks. Given $x \in D \setminus P$, join x to z_0 by a polygonal arc γ in the region D and let z be its first point on P (interior to P, since the endpoints of P are on C). If $z = z_0$, the part of γ just before z_0 lies in one half-disk, joined to z_L or z_R inside $B \setminus P$. Otherwise take the compact subarc K of P from z to z_0 and a strip $N \subseteq D$ around K with $N \setminus P = N_L \sqcup N_R$ (Lemma 2.1); note the strip removes all of P, not only K. The point of γ just before z lies in one track; a small disk at z_0 has its two components inside the two tracks, so that track reaches one half-disk of B. In all cases x is joined in $D \setminus P$ to z_L or z_R, so there are at most two components. □

Theorem 6.2 (Crosscut theorem)**.** *A polygonal crosscut P of $D = \mathrm{Int}(C)$ with endpoints p, q splits D into exactly two regions, namely $\mathrm{Int}(A_1 \cup P)$ and $\mathrm{Int}(A_2 \cup P)$, where A_1, A_2 are the two arcs of C from p to q; and $\mathbb{R}^2 \setminus (C \cup P)$ has exactly three regions, with boundaries C, $A_1 \cup P$, $A_2 \cup P$.*

Proof. Each $J_i = A_i \cup P$ is a Jordan curve, and all three of C, J_1, J_2 are separating by Theorem 5.5; so Lemma 3.4 applies with $\Omega = D$. Here $\Omega^\dagger = \mathrm{Ext}(C)$ is connected, unbounded and disjoint from J_i, hence lies in $\mathrm{Ext}(J_i)$, so the lemma's V_i is $\mathrm{Int}(J_i)$: the two regions $\mathrm{Int}(J_1), \mathrm{Int}(J_2)$ are distinct components of $D \setminus P$, and by Lemma 6.1 they are the only ones. Finally $\mathrm{Ext}(C)$ is a region disjoint from $C \cup P$ whose boundary C misses $\mathbb{R}^2 \setminus (C \cup P)$, so it is the third component (Lemma 1.4); the boundaries are those given by Theorem 5.5 for C, J_1, J_2. □

Lemma 6.3 (Accessible endpoints can be joined)**.** *Let F be a region and let $a \neq b$ be points of ∂F each accessible from F by a polygonal arc meeting ∂F only at its endpoint. Then there is a polygonal crosscut of F from a to b.*

Proof. Join the inner endpoints of the two access arcs by a polygonal arc in the region F (Lemma 1.5). The union meets ∂F only in $\{a, b\}$, and a, b have degree one in its subdivision graph; extract the simple polygonal path from a to b (Lemma 1.5). □

Theorem 6.4 (Square extension)**.** *With $Q = [-1, 1]^2$ and $S = \partial Q$: every homeomorphism $u : C \to S$ extends to a homeomorphism $\overline{F} : C \cup \mathrm{Int}(C) \to Q$.*

Proposition 6.5 (Reduction to the square)**.** *Theorem 6.4 implies that every homeomorphism $f : C \to C'$ extends to a homeomorphism $C \cup \mathrm{Int}(C) \to C' \cup \mathrm{Int}(C')$.*

Proof. Choose $h : C' \to S$ (Lemma 5.2), extend $h \circ f$ and h to H, K by Theorem 6.4, and take $K^{-1} \circ H$. $\square$

The next five sections prove Theorem 6.4. Fix C, u, and write $D = \mathrm{Int}(C)$, $\overline{D} = C \cup D$.

Strong accessibility. A point $p \in C$ is *strongly accessible* if some open disk $B(q,r) \subseteq D$ has $\|q-p\| = r$. If $q \in D$ and a is a nearest point of C, then $B(q, \|q-a\|)$ misses C and, being connected, lies in D: nearest points are strongly accessible. If $B(q,r)$ witnesses p and $v = (q-p)/r$, then for every unit w with $\langle v, w \rangle > \frac{1}{2}$ and $0 < t < r$,

$$\|p + tw - q\|^2 = r^2 + t^2 - 2\,t\,r\,\langle v, w \rangle < r^2 + t(t-r) < r^2,$$

so a whole open cone of short straight segments at p enters $B(q,r) \subseteq D$. Taking a nearest point $a(q) \in C$ for every $q \in \mathbb{Q}^2 \cap D$ gives a countable set A of strongly accessible points, dense in C because $C = \partial D$ (Theorem 5.5) gives rational q with $\|q-p\| < \varepsilon/2$ and then $\|a(q) - p\| \leq 2\|q-p\| < \varepsilon$. Fix this A.

7 Matched cellulations

Proposition 7.1 (Initial pair)**.** *Subdivide C at six boundary vertices: the four u-preimages of the corners of Q, and two further points $a, b \in A$ chosen in two nonadjacent corner arcs, so that $u(a), u(b)$ lie in the relative interiors of opposite sides of Q; subdivide S correspondingly. There is a polygonal crosscut P of D from a to b, and the straight chord $P' = [u(a), u(b)]$ is a crosscut of Q°. The crosscut theorem splits $\overline{D}$ and Q into two closed cells each, and the sides correspond by $\overline{R_i} \cap C = A_i \leftrightarrow \overline{R_i'} \cap S = u(A_i)$, where A_1, A_2 are the two arcs of C from a to b, each realized as a path of boundary edges.*

Proof. The dense set A meets both chosen open corner arcs, even after any finite exclusions. Short straight access segments at a, b exist by strong accessibility, and Lemma 6.3 gives P. Since $u(a), u(b)$ are interior to opposite sides, the interior of $[u(a), u(b)]$ lies in Q° by convexity. Apply Theorem 6.2 on both sides (for the square, its polygonal case). Each skeleton — a six-vertex cycle plus a chord — is 2-connected, and its open inner part, the open chord, is connected. $\square$

A *cellulation* of $\overline{D}$ is a finite partition of $\overline{D}$ into open *cells* — points (*vertices*), open arcs (*edges*), and regions (*faces*) — obtained from the decomposition of Proposition 7.1 by finitely many applications of the two constructors below; a cellulation of Q is the same with S in place of C. The initial cellulation consists of the six boundary vertices, the six outer edges, the crosscut, and the two faces. The constructors are:

(i) *edge subdivision*: choose an interior point of an edge e and replace e by that point and the two edges it cuts e into;

(ii) *face split*: choose a face R and two distinct vertices on its boundary, insert a crosscut of R joining them, and replace R by the crosscut's interior and the two regions the crosscut cuts off.

Cells contained in C (resp. S) are *outer*, the rest *inner*; inner edges are always polygonal arcs with interiors in D (resp. Q°). The *carrier* of a point is the cell containing it. Write $\sigma \preceq \tau$ ("σ is a subcell of τ") when $\sigma \subseteq \overline{\tau}$, and let the *closed star* $\mathrm{St}(\sigma)$ be the union of the closures of the supercells of σ.

A *matched pair* is a cellulation of $\overline{D}$ together with one of Q built by the *same* sequence of constructors, and a homeomorphism g of the two 1-skeletons which restricts to u on C and carries every vertex and edge to its counterpart. To apply a constructor to a matched pair is to apply it on both sides at once: a subdivision inserts corresponding points (via g, hence via u on outer edges); a split inserts crosscuts of corresponding faces between corresponding vertices, g is extended over them by any homeomorphism fixing the endpoints, and the two new faces on each side are matched by their boundary paths, which correspond.

Three lemmas record what this arrangement provides. The first is geometry and holds in either realization separately; the second is bookkeeping; the third is the reason for the whole construction.

Lemma 7.2 (Cell geometry)**.** *In any cellulation:* (a) *the open cells partition the closed domain, and every closed cell is the disjoint union of the open cells it contains;* (b) (*frontier*) *an open cell that meets the closure of a cell is contained in it;* (c) *every face is the interior of a Jordan curve assembled from closed cells, its* boundary cycle*; every boundary cycle contains an inner edge; every outer edge lies on exactly one boundary cycle;* (d) *distinct faces are incomparable under* $\preceq$.

Proof. Induction along the constructors. At the initial stage the crosscut theorem supplies the closure identities $\overline{R_i} = R_i \cup P \cup A_i$, where A_i is the union of the closed boundary edges of the ith path: the cells below R_i are those of that path and of the crosscut, the cells below an edge are its two endpoints, and the two paths share only a and b. This gives (a)–(c).

An edge subdivision replaces $\overline{e}$ by $\overline{e_1} \cup \overline{e_2}$, meeting at the new vertex. The only old closed cells containing a new cell are $\overline{e}$ and, by the old frontier property, the $\overline{\tau}$ with $e \preceq \tau$; this gives (a) and (b), nothing two-dimensional changes, and a boundary cycle that had an inner edge retains an inner subedge of it.

A face split applies Theorem 6.2 inside the Jordan region supplied by (c). The resulting identities $\overline{V_i} = V_i \cup P \cup B_i$ give (a)–(c) for the new cells; each new boundary cycle contains the inserted crosscut, which is an inner edge; and every outer edge of the split face lies on exactly one of the two boundary paths B_i.

For (d), suppose $R \preceq T$ for distinct faces R, T. By (a) the two are disjoint, so $R \subseteq \overline{T} \setminus T = \partial T$. But every point of the Jordan curve ∂T is a limit of points of T, so the nonempty open set R would meet T, contradicting (a). □

Lemma 7.3 (Order, parents, stars)**.** (a) $\preceq$ *is a partial order.* (b) *Every cell produced by a constructor lies in the closure of a unique* $\preceq$*-least old cell, its* parent*: the subdivided edge for its three new cells, the split face for the crosscut's interior and the two new faces, and each unchanged cell for itself.* (c) *A closed cell meets C if and only if it has an outer subcell.*

Proof. (a) Reflexivity and transitivity are immediate. For antisymmetry, let $\sigma \neq \tau$ satisfy $\sigma \preceq \tau \preceq \sigma$; being disjoint, each lies in the other's closure minus itself. If σ were a vertex, τ would lie in $\overline{\sigma} \setminus \sigma = \varnothing$; if σ were an edge, τ would lie in its two endpoints and so be a vertex, which the previous case excludes. By symmetry both are faces, and Lemma 7.2(d) forbids that.

(b) For a subdivision this is the list of containing old closed cells displayed in the previous proof, of which $\overline{e}$ is least. For a split, each new cell lies in the open set R; a closed old vertex or edge is by Lemma 7.2(a) a union of open cells other than R, hence disjoint from R; and an old face $\overline{T}$ containing a new cell forces $R \subseteq \overline{T}$ by the frontier property and then $T = R$ by incomparability. So $\overline{R}$ is the only old closed cell in question.

(c) Every point of C has an outer carrier, since inner edges and vertices lie in D and faces are disjoint from C. So a closed cell meeting C contains a point whose carrier is outer, and the frontier property makes that carrier a subcell. The converse is immediate. □

Lemma 7.4 (Correspondence)**.** *Let $\sigma \mapsto \sigma'$ be the cell bijection of a matched pair. Then $\sigma \preceq \tau$ if and only if $\sigma' \preceq \tau'$; σ is outer if and only if σ' is; and parents correspond. Consequently $\overline{\sigma}$ meets C exactly when $\overline{\sigma'}$ meets S, and $\mathrm{St}(\sigma) \cap \mathrm{St}(\tau) \neq \varnothing$ exactly when $\mathrm{St}(\sigma') \cap \mathrm{St}(\tau') \neq \varnothing$.*

Proof. Each constructor creates its new cells together with their subcell relations and their outer status by the same formulas on both sides — the closure identities in the proof of Lemma 7.2 are literally identical in the two realizations — and the parent rule of Lemma 7.3(b) is combinatorial. Since the two cellulations are built by the same sequence of constructors, induction along it propagates all three correspondences. The first consequence is then Lemma 7.3(c), whose criterion mentions only $\preceq$ and outerness. For the second, a point of $\mathrm{St}(\sigma) \cap \mathrm{St}(\tau)$ lies in $\overline{\alpha} \cap \overline{\beta}$ for some $\alpha \succeq \sigma$ and $\beta \succeq \tau$, and its carrier is, by the frontier property, a common subcell of α and β; the corresponding cell is nonempty and witnesses the intersection on the other side. □

Parents compose along a sequence of refinements. By Lemma 7.3(b) the parent of the carrier of a point is its carrier at the previous stage, and the star of a cell is contained in the star of its parent; so the closed stars of a fixed point shrink, weakly, under refinement.

One caution. During the finite-transfer induction of the next section, intermediate stages may temporarily have a disconnected inner part $|\Gamma| \setminus C$ — after an ear with both endpoints on the outer cycle, for instance. None of the three lemmas above assumes otherwise. A *matched cellulation* is a matched pair whose two skeletons are moreover 2-connected with connected inner part; the constructions below verify this of their final outputs, not of every intermediate stage.

8 Finite transfer

Everything now turns on being able to perform a refinement on one side and reproduce it on the other. In the direction source-to-target this is unobstructed, because the target skeleton is polygonal everywhere, including on S. In the direction target-to-source it is not: a new edge that reaches C must land at a point one can actually reach from inside, which is why the anchors are taken from the strongly accessible set A, and why each of them is given exactly one spoke.

Lemma 8.1 (Local sectors and face access)**.** *Let x be a point of a finite polygonal plane graph. Small closed disks at x meet the graph in finitely many straight radial segments of pairwise distinct directions (the local edge branches at x, whether x is a vertex, a bend, or interior to a straight piece), so the open disk minus the graph is a union of open sectors. Consequently: a face F of one of our cellulations is polygonally accessible at each boundary point x off C; on the target side, at every boundary point including points of S.*

Proof. Segments and vertices not through x are avoided by Lemma 1.2(c); segments through x meet a small disk in radial segments, of distinct directions since edge interiors are disjoint, avoid vertices, and edges are injective. For access on the source side, apply this to the polygonal inner skeleton and shrink the disk off the compact set C; the disk then meets the whole skeleton in the radial segments. Since F accumulates at $x \in \partial F$ and meets the disk only inside sectors, some sector meets F; being connected and disjoint from the skeleton it lies in a single face by Lemma 7.2(a), namely F, and a short straight segment from x into it is an access arc. On the target the whole graph is polygonal; at $x \in S$ some sectors lie outside Q and belong to no face, but the sector meeting F is connected, misses S, and meets Q°, hence stays in Q° (Theorem 3.2 for the square) and lies in F as before. □

Theorem 8.2 (Finite transfer)**.** *Let (Γ, Γ') be one of our matched cellulations.*

(a) *If $H \supseteq \Gamma$ is a finite 2-connected plane graph with outer cycle C, all new edges polygonal with interiors in D, and $|H| \setminus C$ connected, then the refinement $\Gamma \to H$ can be reproduced in the target: there is a matched cellulation extending (Γ, Γ') whose source skeleton is H.*

(b) *Symmetrically from the target side, provided every new edge meeting S has exactly one endpoint on S, at a fresh point of $u(A)$ that becomes a boundary vertex with exactly one incident new edge.*

Proof. First pass to a common subdivision so that the old skeleton is literally a subgraph of the new graph, transferring each new point on an old edge to the other realization by the edge parametrization (u on outer edges); in particular every new boundary vertex — in (b), each fresh point $u(a)$ — is inserted into the outer cycle first. By Lemma 4.1(e), the new graph is then obtained by a finite sequence of ears, and we transfer them one at a time; each ear insertion is at most two subdivisions plus one face split, so all intermediate stages satisfy Lemmas 7.2–7.4, whether or not their inner parts are momentarily disconnected.

The interior of the next ear is connected and disjoint from the current skeleton, hence lies in one current face F (Lemma 7.2(a)), whose boundary is a Jordan curve containing the ear's endpoints; the ear is a crosscut of F. Let F^* be the corresponding face on the other side, and let v^*, w^* be the points corresponding to the ear's endpoints. In case (a) these lie on the skeleton off S or on S as images of boundary vertices already present, and are accessible from F^* by Lemma 8.1. In case (b), an endpoint of the reproduced crosscut is either an inner skeleton point, accessible by Lemma 8.1, or a fresh anchor $a \in A$: there, the open cone of straight access segments furnished by strong accessibility can be shrunk, using Lemma 1.2(c), until its short segments avoid the compact remainder of the skeleton; the segment then lies in the unique current face incident with a. That face is F^*: when a was inserted it was incident with exactly one face, because each outer edge lies on exactly one face boundary (Lemma 7.2(c)), and whenever a later ear split the face incident with a, exactly one of the two boundary paths contained a; the correspondence preserves this face throughout, and on the target side the spoke at $u(a)$, the unique new edge at $u(a)$, is inserted precisely into the corresponding face.

Now Lemma 6.3 joins v^* and w^* by a crosscut of F^*; in case (b) one takes the access segments just constructed. (The crosscut automatically avoids all existing edges, since the open face is disjoint from the skeleton.) Split both faces by the two corresponding crosscuts (Theorem 6.2 on each side) and extend g over the ear by any homeomorphism

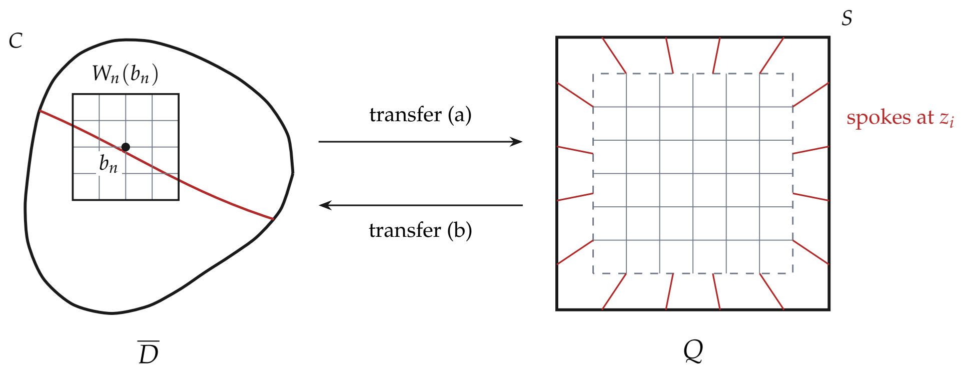


Figure 7: One stage of the scheme. A grid fine enough to force cells below ε_n is attached to the source inside the square $W_n(b_n)$ and carried to the target; then an anchored mesh — a gridded core λQ and a collar cut into rectangles by spokes dropped from fresh anchor points $z_i \in u(A)$ — is laid over the target and carried back. Refining one side alone would shrink that side's cells only; the alternation shrinks both.

fixing the ends. After the last ear, the prescribed-side skeleton is the given graph and the other side is its reproduced counterpart, which is 2-connected because it grew by subdivisions and ears from the 2-connected old skeleton (Lemma 4.1(d)); and the given graph's connected inner part transfers to the reproduced side by the correspondence of incidences (Lemma 7.4): the inner parts of the two skeletons are homeomorphic via g, which matches C with S. Hence the final pair is a matched cellulation. $\square$

9 The refinement scheme

Let $B = \mathbb{Q}^2 \cap D$, let $b_1, b_2, \ldots$ list B with every point occurring infinitely often, and put $\varepsilon_n = 2^{-n}$. For $p \in D$ let $r(p) = \mathrm{dist}_\infty(p, C)$; the open square of radius $r(p)$ at p lies in D, and r is 1-Lipschitz for $\|\cdot\|_\infty$. Put $s_n(p) = r(p) - \min(r(p)/2, \varepsilon_n)$ and let $W_n(p)$ be the closed square of radius $s_n(p)$ at p, so $W_n(p) \subseteq D$ and $r(p) - s_n(p) \leq \varepsilon_n$.

Stage n starts from $(\Gamma_{n-1}, \Gamma'_{n-1})$ and performs four steps (Figure 7): attach a local grid in $W_n(b_n)$ to the source (Proposition 9.1); transfer it to the target (Theorem 8.2(a)); refine the whole target by the anchored mesh with $\delta = \varepsilon_n$ (Proposition 9.2); transfer back (Theorem 8.2(b)). The result is (Γ_n, Γ'_n).

Proposition 9.1 (Local grid)**.** *There is a 2-connected extension $H_n \supseteq \Gamma_{n-1}$, polygonal off C, with $|H_n| \setminus C$ connected, containing a full rectangular grid on $W = W_n(b_n)$ whose lines belong to the skeleton and whose closed rectangles have diameter $< \varepsilon_n$.*

Proof. Let K be such a grid on W (mesh below $\varepsilon_n/2$), subdivided against the inner skeleton of Γ_{n-1} (Lemma 4.1(f)). If the two graphs share at least two vertices, their union is 2-connected (Lemma 4.1(c)). If they share none, K lies in one face F (a bounded Jordan region); take a straight grid edge J with supporting line ℓ: the component of $\ell \cap F$ containing J° is a bounded open segment of ℓ whose closure E is a segment with two distinct endpoints on ∂F and interior in F — a crosscut — and after adding the ear E (Lemma 4.1(d)) the graphs share two vertices along J. If they share exactly one vertex

x, the connected set $K \setminus \{x\}$ lies in one face; run the same line construction through a grid-edge point $y \neq x$, along a line missing x, giving a crosscut E with x, y as two common points. In each case the union, overlaid, is 2-connected and contains the grid.

Its inner part may be disconnected (an auxiliary crosscut may have both ends on C). While it is, pick points in two components, join them by a simple polygonal arc in the region D (Lemma 1.5), and overlay: each component of the arc outside the graph is an ear; ears preserve 2-connectivity, no ear creates a new component of the inner part since the arc lies in D, and once all its ears are present the whole arc joins the two chosen components inside the inner part. Each round strictly lowers the number of components, so finitely many rounds finish. □

Proposition 9.2 (Anchored mesh). *For every $\delta > 0$ there is a polygonal cellulation T of Q whose faces have diameter $< \delta$, whose skeleton is 2-connected with $|T| \setminus S$ connected, in which every prescribed boundary vertex of Γ'_{n-1} appears, and whose only new edges meeting S are single* spokes*, one at each of finitely many fresh points of $u(A)$.*

Proof. Radial coordinates $x = tz$ ($t = \|x\|_\infty$, $z \in S$) identify the collar $Q \setminus (\lambda Q)^\circ$ with $S \times [\lambda, 1]$; fix $\lambda < 1$ with $\sqrt{2}(1-\lambda) < \delta/4$. Parametrize S by a circle, partition it into arcs of image-diameter $< \delta/8$ (uniform continuity), and choose one fresh point z_i of the dense set $u(A)$ in the relative interior of each, avoiding the finitely many excluded points — possible since a dense set meets every open subarc even after finitely many deletions. Consecutive z_i then bound arcs $A_i \subseteq S$ of diameter $< \delta/4$; insert the old boundary vertices as subdivisions of these arcs. Draw the spokes $[z_i, \lambda z_i]$ — disjoint, since two radial segments of the collar meet only on a common ray — cutting the collar into the rectangles $R_i = \{tz : z \in A_i,\ \lambda \le t \le 1\}$, each of diameter at most $2\sqrt{2}(1-\lambda) + \delta/4 < \delta$ by the estimate $\|tz - sw\| \le \|tz - z\| + \|z - w\| + \|w - sw\|$. Grid the core square λQ with mesh $< \delta/2$ so that its rectangles have diameter $< \delta$, and overlay everything. The faces are the grid rectangles and the collar rectangles; the skeleton is 2-connected by Lemma 4.1(c): start from the core grid, subdivided at all the points λz_i, which is 2-connected; each collar-rectangle boundary ∂R_i is a cycle — two spokes, one boundary arc, one core arc — sharing at least the two vertices $\lambda z_i, \lambda z_{i+1}$ with the graph built so far, so adjoining these cycles one after another preserves 2-connectivity, and their union with the grid is the whole skeleton. And $|T| \setminus S$ is connected: the core grid and $\partial(\lambda Q)$ form a connected set in Q°, to which each spoke minus its outer endpoint is attached. Overlaying T with Γ'_{n-1} preserves 2-connectivity, since both graphs contain the outer cycle S and so share at least two vertices (Lemma 4.1(c)); if the two inner parts are disjoint, adding the ears of one joining arc in Q°, as in Proposition 9.1, connects them. This yields the required refinement; the only new edges meeting S are the spokes. □

Proposition 9.3 (Shrinking stars). *For every $x \in D$ one has* $\operatorname{diam} \operatorname{St}_{\Gamma_n}(x) \to 0$, *and* $\operatorname{diam} \operatorname{St}_{\Gamma'_n}(\sigma'_n(x)) \to 0$ *holds uniformly in x; here $\sigma_n(x)$ is the carrier of x and $\sigma'_n(x)$ the corresponding target cell.*

Proof. Uniformity on the target: the stage-n target skeleton contains the mesh of Proposition 9.2, so each of its faces, being connected and disjoint from the mesh skeleton, lies in one mesh face and has diameter $< \varepsilon_n$; faces only shrink under later refinement (each face lies in the closure of its parent), and all closed faces in a star share a point, so star diameters are $< 2\varepsilon_n$.

On the source, fix x and $d = r(x) > 0$; choose $b \in B$ with $\|b - x\|_\infty < d/8$, occurring at arbitrarily large indices n with $\varepsilon_n < d/8$. Then $r(b) > 7d/8$, so $s_n(b) \geq r(b) - \varepsilon_n > 3d/4 > d/8$, and x lies in the interior of $W_n(b)$. At the stage-n pullback, every face of Γ_n incident with x has, by parent compatibility, a parent face T in the pre-pullback source cellulation with $x \in \overline{T}$; T is connected and disjoint from that skeleton, which contains all grid lines of W and ∂W, so T lies in one component of their complement, and not in the unbounded one, whose closure misses the interior point x; hence T lies in one open grid rectangle and $\overline{T}$ in its closure, of diameter $< \varepsilon_n$. All closed faces of $\mathrm{St}_{\Gamma_n}(x)$ contain x, so $\operatorname{diam} \mathrm{St}_{\Gamma_n}(x) < 2\varepsilon_n$; and stars only shrink at later stages. $\square$

Let $A_\infty \subsetneq A$ be the preimages of all fresh anchors ever used. Since stage-n anchors cut S into arcs of diameter $< \varepsilon_n/4$, the set $u(A_\infty)$ is dense in S; and A_∞ is dense in C because u^{-1} is a continuous surjection. When an anchor is introduced its spoke is an inner edge at it, and an initial subedge of the spoke remains incident at all later stages.

10 The interior homeomorphism

Everything is now in place. For $x \in D$ let $T_n(x)$ be the closed target star of $\sigma'_n(x)$. Parent compatibility makes these nested, and Proposition 9.3 drives their diameters to zero, so by Lemma 1.3 their intersection is a single point. That point is the image of x:

$$\{F(x)\} \;:=\; \bigcap_{n\geq 1} T_n(x).$$

Write $G_n = |\Gamma_n|$ and $G'_n = |\Gamma'_n|$; the skeleton homeomorphisms g_n are nested and combine to a bijection $g_\infty : \bigcup_n G_n \to \bigcup_n G'_n$.

Proposition 10.1 (The interior map). *$F : D \to Q^\circ$ is a homeomorphism, and $F = g_n$ on $G_n \cap D$ for every n.*

Proof. Throughout, we use: for any point y of a closed domain, the complement of the union of the closed cells not containing y is a relative neighborhood U of y in which every carrier is a supercell of the carrier of y (a carrier whose closure missed y would be excluded; then the frontier property applies). Call this the *star neighborhood* at y.

Agreement. For $x \in G_N \cap D$ and $n \geq N$, the skeleton map carries the carrier of x onto the corresponding cell, so $g_N(x) = g_n(x) \in T_n(x)$; hence $F(x) = g_N(x)$.

Continuity. Given x and η, pick n with $\operatorname{diam} T_n(x) < \eta$; in the stage-$n$ source star neighborhood at x, shrunk into D, every point z has carrier above that of x, the corresponding target relation holds (Lemma 7.4), so $T_n(z) \subseteq T_n(x)$ and $\|F(z) - F(x)\| < \eta$.

Interior image. Choose n with the source star of x smaller than $\operatorname{dist}(x, C)$, hence disjoint from C; the corresponding target star misses S (Lemma 7.4), so $F(x) \in Q^\circ$.

Injectivity. If $F(x) = F(y)$, the target stars of x, y meet at every stage, hence so do the source stars (Lemma 7.4); a common point gives $\|x - y\| \leq \operatorname{diam} \mathrm{St}_{\Gamma_n}(x) + \operatorname{diam} \mathrm{St}_{\Gamma_n}(y) \to 0$.

Surjectivity. Skeleton points off S are dense in Q°: every face has small diameter and its boundary contains an inner edge. Given $y \in Q^\circ$, take skeleton points $y_k \to y$ off S and $x_k = g_\infty^{-1}(y_k) \in D$. Choose n with the target star of y off S, and its target star neighborhood V; let K be the union of the source closed cells corresponding to the finitely many supercells of the carrier of y: by Lemma 7.4, K is a compact subset of D. For large k, $y_k \in V$, and viewing x_k, y_k as corresponding skeleton points at a late common stage, their stage-n carriers correspond; so $x_k \in K$. The sequence x_k has a limit point $x \in K \subseteq D$ along a subsequence, and by continuity and agreement $F(x) = \lim y_k = y$.

Cells map to cells. For corresponding inner open cells: on vertices and edges $F = g_n$. For a face R: by Lemma 7.2(d) the closed star of R is $\overline{R}$ itself, so $F(R) \subseteq \overline{R'}$, and indeed $F(R) \subseteq R'$: a value on $\partial R'$, which lies in the target skeleton and in Q°, has a source-skeleton preimage in D at which F agrees with g_n, contradicting injectivity. For the converse, a preimage in D of a point of R' cannot have a vertex or edge carrier (its image would be a skeleton point), so its carrier is a face R_0 with $F(R_0) \subseteq R'_0$ meeting R'; faces are disjoint, so $R_0 = R$. Thus $F(R) = R'$.

Inverse continuity. Given $y = F(x)$ and η, pick n with the source star of x smaller than η and take the target star neighborhood V at y: for $z \in V \cap Q^\circ$, the target carrier of y is below that of z; transferring and applying the previous paragraph, the source carrier of x is below that of $F^{-1}(z)$, so $F^{-1}(z)$ lies in the source star of x and $\|F^{-1}(z) - x\| < \eta$. □

11 Continuity at the Jordan curve, and the square extension

Extend F to $\overline{F} : \overline{D} \to Q$ by $\overline{F} = u$ on C; it is a bijection onto Q, continuous on D, and we must prove continuity at each $p \in C$.

Lemma 11.1 (Skeleton crosscuts). *For distinct $a, b \in A_\infty$ in a common finite stage, there is a crosscut P of D from a to b inside the stage's skeleton, whose image $P' = g(P)$ is a crosscut of Q° from $u(a)$ to $u(b)$.*

Proof. Write $X = |\Gamma| \setminus C$, a connected set. If some inner edge has both endpoints on C, its interior is clopen in X, hence all of X, and a, b, being incident with inner edges (anchors), are its endpoints: take that edge. Otherwise let Λ be the vertices and edges entirely in D; attaching to each of its components the half-open interiors of edges with one endpoint on C partitions X into clopen pieces, one per component, so Λ is connected (and nonempty, via the edge at a). The union of Λ with the two edges at a, b is then connected and meets C exactly in $\{a, b\}$; a simple graph path from a to b inside it is the crosscut. Its image under the skeleton homeomorphism is simple, meets S exactly at $u(a), u(b)$, and is polygonal in Q°. □

Lemma 11.2 (Sides correspond). *With P, P' as above, label the components of $D \setminus P$ by $\overline{U_i} \cap C = A_i$ and of $Q^\circ \setminus P'$ by $\overline{U'_i} \cap S = u(A_i)$ (Theorem 6.2). Then $F(U_i) = U'_i$.*

Proof. F is a homeomorphism agreeing with g on $P \cap D$, so it maps the components of $D \setminus P$ onto those of $Q^\circ \setminus P'$; only the labels need identifying. Pick $c \in A_\infty$ in the relative interior of A_i, at a stage containing P, c, and an inner edge e at c. Since $\overline{U_{3-i}} \cap C = A_{3-i}$ misses c, a short initial subarc J of e, with c removed, avoids $\overline{U_{3-i}}$ and P, hence lies in U_i; likewise its image $g(J)$, short enough, lies in U'_i. As $F(J) = g(J)$, the labels match. □

Proposition 11.3 (Boundary continuity). *$\overline{F}$ is continuous, hence a homeomorphism $\overline{D} \to Q$;*

this proves Theorem 6.4.

Proof. Suppose $x_k \to p \in C$ in D but $\overline{F}(x_k) \not\to u(p)$. Passing to a subsequence in the compact set Q, $F(x_k) \to q \neq u(p)$; and $q \in S$, since a limit in Q° would force $x_k \to F^{-1}(q) \in D$. Let $r = u^{-1}(q) \neq p$. By density of A_∞, choose $a, b \in A_\infty$, distinct from p, r, separating them on C; the stages being nested, a, b share a finite stage, giving crosscuts P, P' (Lemma 11.1). Since $p \notin P$ and the closure of the non-p side meets C only in the arc not containing p (Theorem 6.2), all x_k near p lie on the p-side of P; by Lemma 11.2 their images lie on the $u(p)$-side of P', whose closure meets S only in the arc of $u(p)$. But $q = u(r)$ lies in the relative interior of the other arc — contradiction. Continuity of the bijection $\overline{F}$ on the compact set $\overline{D}$ makes it a homeomorphism onto Q. For sequences approaching p within C, continuity is that of u; mixed sequences follow by combining. □

12 From the square extension to the plane

Only the exterior is left. Inversion in a point of $\mathrm{Int}(C)$ turns the closed exterior into a closed interior punctured at one point, so the theorem just proved applies to it — provided the two extensions can be made to agree at that point, which is what the pointed version below arranges.

Lemma 12.1 (Moving a point of the square). *For interior points x, y of Q there is a homeomorphism of Q fixing S pointwise and sending x to y.*

Proof. The four triangles on x and the sides of Q triangulate Q; map each affinely to the triangle on y and the same side, fixing the two side vertices and sending $x \to y$. Adjacent maps agree on shared segments (affine maps agreeing at two points agree on their line), and each fixes its side of S pointwise; paste, and construct the inverse the same way with the roles of x, y reversed. □

Proposition 12.2 (Pointed extension). *Given $f : C \to C'$ and interior points $z \in \mathrm{Int}(C)$, $z' \in \mathrm{Int}(C')$, some extension of f to the closed interiors sends z to z'.*

Proof. Choose a homeomorphism $C' \to S$ (Lemma 5.2) and extend it to $J : C' \cup \mathrm{Int}(C') \to Q$ (Theorem 6.4); let H be an extension of f from Proposition 6.5. Insert the mover (Lemma 12.1) taking $J(H(z))$ to $J(z')$: $J^{-1} \circ M \circ J \circ H$ extends f and sends $z \mapsto z'$. □

Lemma 12.3 (Inversion). *For $a \in \mathrm{Int}(C)$, the inversion $I_a(x) = a + (x - a)/\|x - a\|^2$ maps C onto a Jordan curve C^a and restricts to a homeomorphism of $C \cup \mathrm{Ext}(C)$ onto $\left(C^a \cup \mathrm{Int}(C^a)\right) \setminus \{a\}$.*

Proof. I_a is an involutive homeomorphism of $\mathbb{R}^2 \setminus \{a\}$, so $C^a = I_a(C)$ is a Jordan curve. Let $U = I_a(\mathrm{Ext}(C)) \cup \{a\}$: open away from a, and open at a because the complement of a large disk maps into a punctured small disk at a; connected, since a lies in the closure of the connected image; and bounded, since $\mathrm{Ext}(C)$ avoids the ball $B(a, \mathrm{dist}(a, C)) \subseteq \mathrm{Int}(C)$. Away from a, boundaries correspond under the homeomorphism, so $\partial U = C^a$ (with a

interior to U). Then U is clopen in the component of $\mathbb{R}^2 \setminus C^a$ containing it, hence equals it; being bounded, that component is $\mathrm{Int}(C^a)$ (Theorem 5.5). Removing a and adding $C \to C^a$ gives the stated restriction. $\square$

Proof of Theorem 1.1. Extend f over the closed interiors to $F_{\mathrm{int}} : C \cup \mathrm{Int}(C) \to C' \cup \mathrm{Int}(C')$ (Proposition 6.5). For the exteriors, pick $a \in \mathrm{Int}(C)$, $b \in \mathrm{Int}(C')$ and set $f^* = I_b \circ f \circ I_a : C^a \to (C')^b$. By Proposition 12.2 extend f^* to H^* over the closed interiors with $H^*(a) = b$; then

$$F_{\mathrm{ext}} = I_b \circ H^* \circ I_a : \; C \cup \mathrm{Ext}(C) \longrightarrow C' \cup \mathrm{Ext}(C'),$$

using Lemma 12.3 on both sides and removing the corresponding points a, b, is a homeomorphism extending $I_b \circ f^* \circ I_a = f$ on C. The maps $F_{\mathrm{int}}, F_{\mathrm{ext}}$ agree on C and are defined on the two closed sets $C \cup \mathrm{Int}(C)$ and $C \cup \mathrm{Ext}(C)$ covering the plane; pasting them, and pasting their inverses on $C' \cup \mathrm{Int}(C')$ and $C' \cup \mathrm{Ext}(C')$, yields the desired homeomorphism of $\mathbb{R}^2$ extending f. $\square$

What the proof used

It is worth recording how little went in. Beyond the metric topology of the plane — compactness, connectedness, uniform continuity — and the density of $\mathbb{Q}^2$, the argument rests on one counting function, three constructions, and a single nonplanarity fact.

The counting function is the crossing parity π_C of Section 3. It is what makes the polygonal count exact: the strip lemma bounds the number of components by two, and π_C, taking both values, shows the bound is attained. The identity $\pi_C = \pi_{J_1} + \pi_{J_2}$ is then the entire content of the polygonal crosscut theorem.

The three constructions are the two-sided strip, which manufactures the sides of a polygonal curve; the chain of squares, which buries a simple arc so thoroughly that its complement never notices it; and the matched cellulation, which replaces the homeomorphism being sought by a sequence of finite ones and lets a limit do the rest.

The nonplanarity fact is that $K_{3,3}$ has no plane drawing. It is invoked exactly twice, and both times to forbid a curve from separating wrongly: once against a curve that separates too little — not at all — and once against one that separates too much, into three regions. Between those two applications lies the whole distance from polygons to arbitrary Jordan curves.

Note

The detailed companion manuscript proves the same theorem along the same route with every inference expanded to formalization granularity, together with a mechanically generated citation index and an explicit imported-background appendix. Where the present text says "by the correspondence of incidences" or compresses a constructor case check into a sentence, the companion contains the full case analysis.